

UNIFIED RESEARCH-LEVEL CURRICULUM

Mathematics, Theoretical Physics, and Computer Graphics

A single dependency-aware programme with one rigorous mathematical spine

Baseline	40 focused hours/week; 48 scheduled weeks/year
Nominal scale	12 years; 48 quarters; exactly 23,040 scheduled focused hours
Architecture	36 mathematics volumes + 15 physics volumes + 17 graphics volumes = 68 retained volumes
Merge rule	Shared foundations appear once. If treatments differ, the proof-level mathematical treatment wins.

40-hour detailed scheduling edition - August 2026

Built from the latest completed Mathematics v6, Theoretical Physics Refined, and Computer Graphics v4 curricula.

1. What this unified curriculum changes

The three completed plans were designed independently, so each correctly repeated some foundations. This master curriculum removes those repetitions without deleting genuine subject-specific depth. Mathematics becomes the sole owner of theorem-level foundations; physics owns physical models, approximation regimes and intuition; computer graphics owns discretization, algorithms, systems implementation and visual-computing research. The result is not a lowest-common-denominator merge. It deliberately chooses the strongest mathematical formulation whenever two routes overlap.

Central principle. Study an object at full mathematical strength once, then reuse it. Do not relearn Banach spaces as a physics shortcut, Fourier analysis as a graphics shortcut, or probability as a Monte Carlo shortcut. Instead, return to the rigorous mathematics volume and apply it in the new domain.

1.1 The three source authorities

Authority	Source plan	What it owns in the merged programme
Mathematics	Master Mathematics Programme v6 - audited, schedule-specific	Definitions, theorem statements, proof standards, counterexamples, abstract structures, research-level mathematical capstones.
Theoretical physics	7-Year Executable Reading Curriculum - Refined	Physical principles, derivations, controlled approximations, physical intuition, model hierarchy, canonical calculations and physics research bridges.
Computer graphics	Research-Level Master Curriculum v4	Continuous-to-discrete translation, programming laboratories, numerical diagnostics, rendering/geometry/simulation systems, paper reproduction and graphics research practice.

1.2 Non-negotiable rigor standard

- Foundational mathematics: reproduce complete proofs, not just proof sketches, for the central theorems designated as foundational in the mathematics plan.
- Advanced mathematics: know the exact statement, hypotheses, examples/counterexamples, dependency graph, full proof architecture, and at least one major technical lemma in full detail.
- Physics: derive canonical equations from stated principles, identify the small parameter or scaling behind each approximation, and know which statements are heuristic, asymptotic, or mathematically rigorous.
- Graphics: derive the governing continuous model where one exists, justify the discretization, implement a reduced reference version, test convergence/stability when meaningful, and separate modeling, discretization, estimator, optimization and software errors.
- Cross-domain claims: when a physics or graphics source uses a mathematically informal statement, replace the statement by the strongest version available in the mathematics curriculum before using it.

2. Deduplication: what is studied once and where

Common topic	Single rigorous owner	What remains in applied tracks
Logic, sets, proof	M01 Proof; M05 Logic	Delete elementary proof introductions in physics and graphics. Keep physics logic only when it concerns interpretation or model assumptions.
Linear algebra	M02 Linear Algebra	Physics keeps operator/state applications; graphics keeps numerical/sparse implementations and transforms, but not a second abstract linear-algebra course.
Calculus and real analysis	M06 Real Analysis	Physics and graphics use it; no independent calculus foundations.
Complex analysis	M08 Complex Analysis	Physics keeps contour methods as applications; graphics uses complex methods only where genuinely useful.
Fourier/harmonic analysis	M09 Harmonic Analysis; M35 modern endpoint	Physics keeps spectral interpretation; graphics keeps sampling/filtering algorithms. Mathematical convergence and operator theory are not retaught.
Measure and probability	M14 Probability; M15 stochastic methods	Physics keeps stochastic models; graphics keeps Monte Carlo estimators. Measure-theoretic probability appears once.
Functional/operator analysis	M07 Functional Analysis	Quantum mechanics/QFT apply the rigorous framework; domain/self-adjointness issues are upgraded rather than simplified away.
Distributions and PDE	M10 PDE	Physics owns the governing PDEs and physical boundary conditions; graphics owns discretization and solvers.
Topology/manifolds/geometry	M11 Topology; M12 Smooth Manifolds; M13 Geometric Topology	GR, gauge theory and geometry processing use these foundations instead of rebuilding them.
Lie groups/representation theory	M18 Lie and Representation Theory	Physics retains symmetry applications; graphics retains transformation-group applications.
Stochastic calculus	M15 SDE/Random Structures	Physics P18 and SLE/particle tracks apply it; no separate elementary Ito course in physics.
Continuum mechanics/PDE rigor	M10 for analysis; P11 for physical continuum model	G10-G11 retain FEM/PIC/FLIP/MPM/contact and solver engineering, not duplicate continuum foundations.
Optics and waves	P07 for physical wave/EM foundations	G06/G12 retain radiometry, rendering, camera/sensor models, computational imaging and displays.
Statistical mechanics / interacting particles	P10 for physics; M17 for rigorous particle/Ising endpoint	Use both layers deliberately: physical theory plus rigorous limits/phase-transition mathematics, without repeating probability basics.
Numerics and optimization	Retained in expanded G02	Original graphics mathematical-foundation volume is dissolved, but numerical linear algebra, conditioning, optimization, autodiff and time-integration practice are preserved here because they are not duplicates of proof-oriented mathematics.

2.1 Volumes removed as standalone units

Physics Volumes I-V and Graphics Volume 1 are not scheduled as independent volumes. Their shared mathematical material is replaced by the corresponding M-volumes. Unique pieces are transferred rather than discarded: asymptotic/perturbative physics methods are attached to P06/P09/P18; numerical conditioning, optimization, automatic differentiation and time integration from the former graphics foundation are folded into G02.

3. The 68-volume architecture

The unified library keeps the original subject identities by prefixing volumes M, P and G. This prevents ambiguous renumbering and makes it easy to return to the detailed source plan for exact week cards, readings and exercises.

Range	Count	Layer	Rule
M01-M36	36	Rigorous mathematics spine	Full v6 sequence retained. This is the prerequisite authority for all other tracks.
P06-P20	15	Theoretical physics	Physics mathematical Volumes I-V removed; all physical volumes retained.
G02-G18	17	Computer graphics	Graphics mathematical Volume 1 dissolved; unique numerical material moved into G02.

3.1 Exact 12-year capacity at 40 h/week

The unified curriculum contains approximately 23,040 focused hours after deduplication. At 40 focused hours/week for 48 scheduled weeks/year, the capacity is exactly 1,920 hours/year and 23,040 hours over 12 years. The 40 mathematics quarters are not compressed: each still receives 30 h/week for 12 weeks (360 hours), exactly matching Mathematics v6. The additional 10 h/week is used for prerequisite-compatible physics, graphics and mathematical animation. After all 40 mathematics quarters are complete, the final eight quarters use the full 40 h/week for advanced physics/graphics/research synthesis.

Period	Scheduled quarters	Weekly structure	Hour accounting
Years 1-10	40 quarters	30 h/week mathematics + 10 h/week applied	14,400 h mathematics + 4,800 h applied
Years 11-12	8 quarters	40 h/week applied/research	3,840 h applied/research
Total	48 quarters	48 weeks/year x 12 years	23,040 focused hours

No rigor compression. The calendar becomes shorter only because the weekly capacity increases from 30 to 40 hours. The 14,400 mathematics hours and the proof/reconstruction standards are unchanged. If the extra 10 hours causes mathematics proof quality to fall, reduce the applied track temporarily rather than taking hours from mathematics.

3.2 Standard weekly operating schedule: Years 1-10

Use a five-day, eight-hour working week. Each day contains six hours of mathematics and two hours of the current applied segment. The exact time of day is flexible; the cognitive order is not: new theory before routine exercises, hard proofs before low-value polishing, and implementation only after the continuous model is understood.

Day	Mathematics: 6 h	Applied: 2 h	Non-negotiable output
Monday	3 h definitions/theorems + 3 h guided proof reading/reconstruction	2 h applied theory/derivation	Build the week's conceptual map; write exact hypotheses before using results.
Tuesday	1 h recall + 5 h exercises/proofs	2 h applied worked problems / numerical derivation	Attack core problems without notes first; maintain a failure log.
Wednesday	3 h second-pass theory + 3 h hard problems	2 h implementation / mathematical animation	Convert theory into examples, counterexamples, code or visual explanation.
Thursday	4 h proof reconstruction + 2 h examples/counterexamples	2 h implementation / validation	Reproduce at least one central argument from memory; test one applied claim numerically.
Friday	3 h cumulative synthesis + 3 h closed-book problems/oral exam	2 h project checkpoint / paper bridge	Finish with a written weekly audit and explicit carry-over list.

3.3 Standard 12-week quarter rhythm: Years 1-10

Weeks	Purpose	Mathematics (30 h/week)	Applied (10 h/week)
Weeks 1-3	Build foundations	Execute Mathematics-v6 weeks 1-3; definitions, main examples, first proof chain.	Applied segment: first conceptual block; reproduce baseline derivations/algorithms.
Week 4	Checkpoint I	Closed-book definitions + one full proof + representative problems. Repair any missing prerequisite immediately.	Applied mini-check: one derivation or executable baseline plus a correctness test.
Weeks 5-7	Core depth	Execute Mathematics-v6 weeks 5-7; hard exercises, counterexamples and structural theorems.	Applied segment: second conceptual block; implementation/problem set grows beyond toy cases.
Week 8	Checkpoint II	Timed problem set and proof reconstruction; identify theorem dependencies explicitly.	Applied diagnostic: compare analytic expectation with numerical/visual result.
Weeks 9-10	Synthesis	Execute Mathematics-v6 weeks 9-10; mixed problems and cross-links to earlier volumes.	Applied segment: third conceptual block; begin mini-project/reproduction.
Week 11	Research bridge	Read one advanced note/paper only after core problems are secure; write a theorem map.	Stress-test project, record failures, and connect the segment to its rigorous mathematical owner.
Week 12	Quarter gate	6-8 h closed-book exam; 1-2 oral proofs; error correction; one-page dependency map.	Validated deliverable + 2-page technical note. Advancement requires both math and applied gates.

3.4 Applied-segment accounting

Each original interdisciplinary A-quarter represented 360 hours. In the 40-hour plan it is split into three 120-hour segments, labelled I, II and III. During Years 1-10 a segment is studied at 10 h/week for 12 weeks. During Years 11-12 a segment is studied intensively for three weeks at 40 h/week. The content therefore receives the same total hours; only the calendar placement changes.

4. Twelve-year detailed chronological schedule

The schedule below is the authoritative 40-hour calendar. Years 1-10 preserve every audited Mathematics-v6 quarter at its original 30 h/week while adding exactly one 120-hour applied segment at 10 h/week. Years 11-12 contain the remaining 32 applied segments as four consecutive 3-week intensives per quarter. This is an hour-preserving transformation of the 30-hour unified plan, not a reduction in content.

Source-week rule. Physics and graphics source plans were written with their own weekly hour assumptions. In this unified calendar, advance through those source week cards by completed hours and mastery, not by calendar-week labels. Never force a 20-hour physics source week into a 6-hour physics allocation. The detailed source PDFs remain the reading/problem/programming manuals; this document controls calendar placement.

Year 1: 1,920 scheduled hours

Y1 Q1 (global Q1) - 480 h

Mathematics - 30 h/week (360 h): M01 - Proof and mathematical language. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A01-1 - Scientific computing foundation; Tooling and numerical discipline. Python/C++ workflow, Git, tests, profiling; floating-point error, conditioning and reproducible experiments; begin Manim scenes for vectors, functions and transformations.

Applied exit evidence. A tested numerical mini-library plus two short mathematical animations.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Y1 Q2 (global Q2) - 480 h

Mathematics - 30 h/week (360 h): M02 - Linear algebra. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A01-2 - Scientific computing foundation; Numerical linear algebra and optimization. Dense/sparse linear algebra, least squares, iterative solves, constrained/unconstrained optimization, autodiff and gradient checking; animate convergence and conditioning failures.

Applied exit evidence. Benchmark notebook/program with condition-number and gradient-check diagnostics.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Y1 Q3 (global Q3) - 480 h

Mathematics - 30 h/week (360 h): M03 - Abstract algebra. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A01-3 - Scientific computing foundation; Time integration, GPU and performance. ODE time integrators, stability intuition, parallel/GPU execution, memory hierarchy, kernels and profiling; package one reusable animation/simulation framework.

Applied exit evidence. GPU or parallel mini-project with profiling report and reproducible build.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Y1 Q4 (global Q4) - 480 h

Mathematics - 30 h/week (360 h): M06 - Real analysis. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A02-1 - Classical mechanics I + image science; Newtonian mechanics + sampled images. Newton laws, conservation, central-force examples and dimensional analysis; image arrays, sampling grids, convolution and reconstruction.

Applied exit evidence. Derive and simulate one mechanical system; demonstrate aliasing on controlled signals/images.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Year-end flexible weeks (Weeks 49-52). Week 49 cumulative mathematics audit; Week 50 physics derivation audit or prerequisite repair; Week 51 code/reproducibility and animation archive; Week 52 recovery or strictly limited backlog. Do not schedule new core content here.

Year 2: 1,920 scheduled hours

Y2 Q1 (global Q5) - 480 h

Mathematics - 30 h/week (360 h): M05 - Mathematical logic. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A02-2 - Classical mechanics I + image science; Lagrangian mechanics + filtering. Action/Euler-Lagrange equations, constraints and small oscillations; Fourier-domain interpretation of filtering, resampling and antialiasing.

Applied exit evidence. Lagrangian derivation set plus an antialiasing/reconstruction implementation.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Y2 Q2 (global Q6) - 480 h

Mathematics - 30 h/week (360 h): M04 - Commutative and homological algebra. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A02-3 - Classical mechanics I + image science; Coupled oscillations + color/perception. Normal modes, driven/damped systems and resonance; color spaces, radiometric vs perceptual quantities, HDR/tonemapping.

Applied exit evidence. Mode-analysis project and color/HDR pipeline with documented assumptions.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Y2 Q3 (global Q7) - 480 h

Mathematics - 30 h/week (360 h): M08 - Complex analysis. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A03-1 - Classical mechanics II + real-time graphics; Hamiltonian structure + raster pipeline. Legendre transform, Hamilton equations, Poisson brackets and canonical variables; clip coordinates, primitives, barycentrics and depth.

Applied exit evidence. Hamiltonian derivations plus CPU reference rasterizer.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Y2 Q4 (global Q8) - 480 h

Mathematics - 30 h/week (360 h): M11 - Topology. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A03-2 - Classical mechanics II + real-time graphics; Canonical mechanics + shading/textures. Canonical transformations, phase portraits and conserved structure; interpolation, texture sampling, mipmapping and local shading.

Applied exit evidence. Symplectic/phase-space experiment plus textured rasterizer.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Year-end flexible weeks (Weeks 49-52). Week 49 cumulative mathematics audit; Week 50 physics derivation audit or prerequisite repair; Week 51 code/reproducibility and animation archive; Week 52 recovery or strictly limited backlog. Do not schedule new core content here.

Year 3: 1,920 scheduled hours

Y3 Q1 (global Q9) - 480 h

Mathematics - 30 h/week (360 h): M14 - Probability. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A03-3 - Classical mechanics II + real-time graphics; Perturbation/kinematics + real-time systems. Perturbative mechanics entry, rigid transforms and kinematics; GPU pipeline, shadows and physically based real-time approximations.

Applied exit evidence. Interactive real-time scene with kinematic animation and performance profile.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Y3 Q2 (global Q10) - 480 h

Mathematics - 30 h/week (360 h): M07 - Functional analysis. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A04-1 - Electromagnetism I + ray tracing; Electrostatics/magnetostatics + ray geometry. Fields, potentials, boundary-value intuition and force laws; rays, camera models, primitive intersections and robust numerical geometry.

Applied exit evidence. Field derivations plus a minimal ray tracer.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Y3 Q3 (global Q11) - 480 h

Mathematics - 30 h/week (360 h): M12 - Smooth manifolds and differential geometry. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A04-2 - Electromagnetism I + ray tracing; Maxwell equations + acceleration structures. Differential/integral Maxwell equations, waves and conservation; BVH/spatial acceleration, materials and recursive transport.

Applied exit evidence. Maxwell derivation notebook plus BVH-accelerated renderer.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Y3 Q4 (global Q12) - 480 h

Mathematics - 30 h/week (360 h): M22 - Elementary and analytic number theory. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A04-3 - Electromagnetism I + ray tracing; Radiation/waves + renderer architecture. Radiation, wave propagation, polarization/coherence awareness; robust scene representation, testing and renderer systems design.

Applied exit evidence. Validated renderer with analytic intersection tests and wave/EM explanatory visualization.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Year-end flexible weeks (Weeks 49-52). Week 49 cumulative mathematics audit; Week 50 physics derivation audit or prerequisite repair; Week 51 code/reproducibility and animation archive; Week 52 recovery or strictly limited backlog. Do not schedule new core content here.

Year 4: 1,920 scheduled hours

Y4 Q1 (global Q13) - 480 h

Mathematics - 30 h/week (360 h): M09 - Harmonic analysis. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A05-1 - Electromagnetism II/SR + light transport I; Special relativity + radiometry. Lorentz transformations, four-vectors and covariant EM entry; radiance/irradiance/solid angle and measurement conventions.

Applied exit evidence. Relativity derivation set plus radiometry unit tests.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Y4 Q2 (global Q14) - 480 h

Mathematics - 30 h/week (360 h): M10 - Partial differential equations. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A05-2 - Electromagnetism II/SR + light transport I; Radiation + BSDFs.

Lienard-Wiechert/radiation structure at appropriate depth; BRDF/BTDF/BSDF properties, reciprocity, energy conservation and material models.

Applied exit evidence. Radiation calculation plus BSDF implementation with conservation checks.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Y4 Q3 (global Q15) - 480 h

Mathematics - 30 h/week (360 h): M18 - Lie groups and representation theory. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A05-3 - Electromagnetism II/SR + light transport I; Covariant EM + rendering equation. Field tensor/covariant formulation and invariant viewpoint; derive rendering equation and implement deterministic/Monte Carlo sanity cases.

Applied exit evidence. Covariant EM note plus rendering-equation solver on benchmark scenes.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Y4 Q4 (global Q16) - 480 h

Mathematics - 30 h/week (360 h): M24 - Algebraic number theory. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A06-1 - Thermodynamics/stat mech + light transport II; Thermodynamic structure + Monte Carlo basics. Thermodynamic potentials, Legendre structure and equations of state; expectation/variance in rendering, direct-light estimators and confidence diagnostics.

Applied exit evidence. Thermodynamic derivations plus unbiased estimator tests.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Year-end flexible weeks (Weeks 49-52). Week 49 cumulative mathematics audit; Week 50 physics derivation audit or prerequisite repair; Week 51 code/reproducibility and animation archive; Week 52 recovery or strictly limited backlog. Do not schedule new core content here.

Year 5: 1,920 scheduled hours

Y5 Q1 (global Q17) - 480 h

Mathematics - 30 h/week (360 h): M25 - Geometry of numbers and Diophantine approximation. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A06-2 - Thermodynamics/stat mech + light transport II; Ensembles/interactions + path tracing/MIS. Microcanonical/canonical/grand-canonical ensembles, partition functions and interacting systems; path tracing and multiple importance sampling.

Applied exit evidence. Partition-function problem set plus MIS path tracer.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Y5 Q2 (global Q18) - 480 h

Mathematics - 30 h/week (360 h): M15 - SDEs and random structures. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A06-3 - Thermodynamics/stat mech + light transport II; Phase transitions + participating media. Order parameters, correlations, mean-field/critical intuition and asymptotic approximations; volume transport, phase functions and transmittance estimators.

Applied exit evidence. Phase-transition computation plus participating-media benchmark.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Y5 Q3 (global Q19) - 480 h

Mathematics - 30 h/week (360 h): M19 - Smooth dynamics, KAM and billiards. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A07-1 - Quantum mechanics I + geometric modeling; QM foundations + parametric curves. Hilbert-space/state/postulate structure, observables and measurement; Bezier curves, B-splines and de Casteljau/de Boor algorithms.

Applied exit evidence. Finite-dimensional QM derivations plus curve-modeling library.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Y5 Q4 (global Q20) - 480 h

Mathematics - 30 h/week (360 h): M21 - Geometric group theory and property (T). Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A07-2 - Quantum mechanics I + geometric modeling; Spectra/spin + surfaces/subdivision. Spectral problems, angular momentum/spin and symmetry; tensor-product surfaces, subdivision schemes and continuity diagnostics.

Applied exit evidence. Spin/spectrum problems plus subdivision surface implementation.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Year-end flexible weeks (Weeks 49-52). Week 49 cumulative mathematics audit; Week 50 physics derivation audit or prerequisite repair; Week 51 code/reproducibility and animation archive; Week 52 recovery or strictly limited backlog. Do not schedule new core content here.

Year 6: 1,920 scheduled hours

Y6 Q1 (global Q21) - 480 h

Mathematics - 30 h/week (360 h): M28 - Additive combinatorics and approximate groups. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A07-3 - Quantum mechanics I + geometric modeling; Exactly solvable systems + implicit geometry. Harmonic oscillator, hydrogenic structure and approximation readiness; implicit/SDF/CSG representations and robust modeling operations.

Applied exit evidence. QM mini-exam plus implicit modeling project with gradient/normal tests.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Y6 Q2 (global Q22) - 480 h

Mathematics - 30 h/week (360 h): M29 - Expanders, spectral gaps and Bourgain-Gamburd. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A08-1 - Quantum mechanics II + DDG/animation; Approximation methods + discrete operators. Perturbation theory/variational/WKB structure; mesh differential operators, cotangent Laplacian and convergence on analytic surfaces.

Applied exit evidence. Approximation-method derivations plus Laplacian convergence study.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Y6 Q3 (global Q23) - 480 h

Mathematics - 30 h/week (360 h): M23 - Advanced analytic number theory. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A08-2 - Quantum mechanics II + DDG/animation; Scattering/open systems + parameterization/geodesics. Scattering and density-matrix/open-system entry; parameterization, geodesics, curvature and discrete energies.

Applied exit evidence. Scattering calculation plus geometry-processing benchmark.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Y6 Q4 (global Q24) - 480 h

Mathematics - 30 h/week (360 h): M27 - Finite fields. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A08-3 - Quantum mechanics II + DDG/animation; Path integrals + deformation/character motion. Path-integral viewpoint and semiclassical link; deformation, vector fields/Hodge ideas, skeletal animation and inverse kinematics.

Applied exit evidence. Path-integral note plus deformation/IK demo with constraint diagnostics.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Year-end flexible weeks (Weeks 49-52). Week 49 cumulative mathematics audit; Week 50 physics derivation audit or prerequisite repair; Week 51 code/reproducibility and animation archive; Week 52 recovery or strictly limited backlog. Do not schedule new core content here.

Year 7: 1,920 scheduled hours

Y7 Q1 (global Q25) - 480 h

Mathematics - 30 h/week (360 h): M26 - Diophantine geometry, transcendence and Hilbert 10. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A09-1 - Continuum/fluid physics + DDG II; Continuum kinematics + variational geometry processing. Deformation gradient, strain, stress and balance laws; mesh energies, weak formulations and smoothing/deformation.

Applied exit evidence. Continuum derivation set plus variational mesh solver.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Y7 Q2 (global Q26) - 480 h

Mathematics - 30 h/week (360 h): M30 - Automorphic forms. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A09-2 - Continuum/fluid physics + DDG II; Euler/Navier-Stokes + advanced DDG. Derive incompressible/compressible equations and nondimensional groups; discrete curvature, exterior/Hodge viewpoints and convergence tests.

Applied exit evidence. Fluid-equation derivation plus DDG convergence report.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Y7 Q3 (global Q27) - 480 h

Mathematics - 30 h/week (360 h): M31 - Algebraic geometry. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A09-3 - Continuum/fluid physics + DDG II; Stability/turbulence entry + robust geometry. Linear stability, vorticity, turbulence phenomenology and scaling; remeshing, topology/robustness and adversarial meshes.

Applied exit evidence. Stability calculation plus robust geometry-processing project.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Y7 Q4 (global Q28) - 480 h

Mathematics - 30 h/week (360 h): M32 - Arithmetic geometry. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A10-1 - Classical fields/GR + simulation I; Classical fields + particles/rigid bodies. Field action/Euler-Lagrange equations and stress-energy ideas; particle systems, rigid-body dynamics and time integration.

Applied exit evidence. Field-theory derivations plus rigid-body simulator.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Year-end flexible weeks (Weeks 49-52). Week 49 cumulative mathematics audit; Week 50 physics derivation audit or prerequisite repair; Week 51 code/reproducibility and animation archive; Week 52 recovery or strictly limited backlog. Do not schedule new core content here.

Year 8: 1,920 scheduled hours

Y8 Q1 (global Q29) - 480 h

Mathematics - 30 h/week (360 h): M36 - Cryptography. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A10-2 - Classical fields/GR + simulation I; Differential geometry of GR + collision/contact. Metric, connection, curvature and geodesics; collision detection, impulses, constraints and contact solvers.

Applied exit evidence. Geodesic/curvature calculations plus contact benchmark suite.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Y8 Q2 (global Q30) - 480 h

Mathematics - 30 h/week (360 h): M16 - Percolation and SLE. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A10-3 - Classical fields/GR + simulation I; Einstein equations + simulation validation. Einstein equation structure, exact-solution examples and causal interpretation; conservation, stability and numerical validation practices.

Applied exit evidence. GR synthesis note plus physically validated simulation case.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Y8 Q3 (global Q31) - 480 h

Mathematics - 30 h/week (360 h): M17 - Particle systems, kinetic limits and Ising. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A11-1 - Condensed matter I + simulation I deepening; Crystals/phonons + deformable solids. Bravais/reciprocal lattices, lattice vibrations and phonons; elasticity models, FEM weak form and element assembly.

Applied exit evidence. Phonon calculation plus small FEM code.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Y8 Q4 (global Q32) - 480 h

Mathematics - 30 h/week (360 h): M20 - Homogeneous dynamics. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A11-2 - Condensed matter I + simulation I deepening; Band theory + cloth/shells/rods. Bloch theorem, bands and semiclassical dynamics; cloth, shells/rods, constraints and stiffness.

Applied exit evidence. Band-structure toy model plus deformable-surface simulator.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Year-end flexible weeks (Weeks 49-52). Week 49 cumulative mathematics audit; Week 50 physics derivation audit or prerequisite repair; Week 51 code/reproducibility and animation archive; Week 52 recovery or strictly limited backlog. Do not schedule new core content here.

Year 9: 1,920 scheduled hours

Y9 Q1 (global Q33) - 480 h

Mathematics - 30 h/week (360 h): M34 - Fractal measures and entropy methods. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A11-3 - Condensed matter I + simulation I deepening; Transport + robust FEM/contact. Boltzmann/semiclassical transport and response; nonlinear/implicit solves, contact coupling and convergence/energy diagnostics.

Applied exit evidence. Transport calculation plus FEM/contact validation report.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Y9 Q2 (global Q34) - 480 h

Mathematics - 30 h/week (360 h): M35 - Modern harmonic and geometric analysis. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A12-1 - Semiconductor foundations + simulation II-a; Semiconductor bands/statistics + Eulerian fluids. Carrier statistics, effective mass and drift/diffusion foundations; grids, advection and incompressibility.

Applied exit evidence. Carrier-statistics calculation plus grid-fluid baseline.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Y9 Q3 (global Q35) - 480 h

Mathematics - 30 h/week (360 h): M13 - Geometric topology, Ricci flow and Hauptvermutung. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A12-2 - Semiconductor foundations + simulation II-a; Poisson-continuity + pressure projection. Self-consistent electrostatics/transport equations; MAC grids, pressure Poisson solve, projection and boundary conditions.

Applied exit evidence. 1D semiconductor PDE solver plus incompressible projection solver.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Y9 Q4 (global Q36) - 480 h

Mathematics - 30 h/week (360 h): M33 - Etale cohomology and Weil/Deligne. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A12-3 - Semiconductor foundations + simulation II-a; Carrier transport + PIC/FLIP/APIC. Mobility/diffusion/recombination modeling; particle-grid transfers and numerical dissipation/momentum diagnostics.

Applied exit evidence. Transport benchmark plus PIC/FLIP/APIC comparison.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Year-end flexible weeks (Weeks 49-52). Week 49 cumulative mathematics audit; Week 50 physics derivation audit or prerequisite repair; Week 51 code/reproducibility and animation archive; Week 52 recovery or strictly limited backlog. Do not schedule new core content here.

Year 10: 1,920 scheduled hours**Y10 Q1 (global Q37) - 480 h**

Mathematics - 30 h/week (360 h): MS37 - Arithmetic synthesis. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A13-1 - Semiconductor devices/quantum transport + simulation II-b; Junctions/MOS + MPM basics. p-n junction electrostatics/current and MOS electrostatics; MPM transfer cycle, constitutive models and grid update.

Applied exit evidence. Device derivations plus minimal MPM implementation.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Y10 Q2 (global Q38) - 480 h

Mathematics - 30 h/week (360 h): MS38 - Groups/dynamics synthesis. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A13-2 - Semiconductor devices/quantum transport + simulation II-b;

Heterostructures/quantum transport + multiphysics. Quantum wells/heterojunctions, Landauer and NEGF entry; coupled solid/fluid/thermal or contact multiphysics patterns.

Applied exit evidence. Quantum-transport toy calculation plus coupled-simulation prototype.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Y10 Q3 (global Q39) - 480 h

Mathematics - 30 h/week (360 h): MS39 - Analysis/probability synthesis. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A13-3 - Semiconductor devices/quantum transport + simulation II-b; Device synthesis + physical cameras. Device scaling/nonidealities and numerical device modeling; pinhole/thin-lens/sensor response, noise and calibration.

Applied exit evidence. Device simulation report plus calibrated camera/image-formation model.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Y10 Q4 (global Q40) - 480 h

Mathematics - 30 h/week (360 h): MS40 - Mathematics research capstone. Execute the corresponding Mathematics v6 12-week schedule without shortening proofs or precision problems. Weeks 4, 8 and 12 use the quarter-gate pattern in Section 3.3.

Applied - 10 h/week (120 h): A14-1 - Many-body I + computational imaging; Second quantization + computational optics. Fock space, creation/annihilation operators and simple many-body Hamiltonians; diffraction/PSF/OTF and sensor image formation.

Applied exit evidence. Second-quantization derivations plus computational-optics simulation.

Microcycle. Weeks 1-4: foundations/baseline; Weeks 5-8: derivation + harder cases; Weeks 9-11: implementation/reproduction + diagnostics; Week 12: validated deliverable and 2-page technical note.

Year-end flexible weeks (Weeks 49-52). Week 49 cumulative mathematics audit; Week 50 physics derivation audit or prerequisite repair; Week 51 code/reproducibility and animation archive; Week 52 recovery or strictly limited backlog. Do not schedule new core content here.

Year 11: full-time applied and research synthesis - 1,920 scheduled hours

Y11 Q1 (global Q41) - 480 h

Applied/research intensive - 40 h/week (480 h): four consecutive 3-week, 120-hour segments. Treat each segment as a short full-time course: Week A theory/derivations, Week B problems/implementation, Week C validation/reproduction/oral gate.

Weeks 1-3: A14-2 - Many-body I + computational imaging; Hartree-Fock + inverse imaging. Mean-field/Hartree-Fock, basis representations and self-consistency; deconvolution, regularization and inverse-problem conditioning. Exit: Self-consistent HF toy code plus inverse-imaging benchmark.

Weeks 4-6: A14-3 - Many-body I + computational imaging; Green functions + displays/sensing. Single-particle Green functions, spectral functions and response entry; display/sensor systems, calibration and computational photography. Exit: Green-function calculation plus end-to-end imaging system analysis.

Weeks 7-9: A15-1 - Many-body II + inverse graphics I; Correlation/collective modes + differentiable rendering basics. Beyond mean field, response and collective excitations; differentiating image formation and gradient verification. Exit: Many-body response problems plus differentiable toy renderer with finite-difference checks.

Weeks 10-12: A15-2 - Many-body II + inverse graphics I; Superconductivity/topological matter + inverse estimation. BCS/order-parameter/topological-band ideas; optimize shape/material/light/pose with priors and regularization. Exit: Superconductivity/topology synthesis plus inverse-rendering recovery benchmark.

Y11 Q2 (global Q42) - 480 h

Applied/research intensive - 40 h/week (480 h): four consecutive 3-week, 120-hour segments. Treat each segment as a short full-time course: Week A theory/derivations, Week B problems/implementation, Week C validation/reproduction/oral gate.

Weeks 1-3: A15-3 - Many-body II + inverse graphics I; Many-body synthesis + identifiability. Quasiparticles, correlation diagnostics and phase structure; visibility discontinuities, ambiguity, identifiability and uncertainty. Exit: Many-body oral exam plus inverse-graphics failure/identifiability report.

Weeks 4-6: A16-1 - QFT foundations + inverse graphics II; Classical fields to quantization + inverse visibility. Canonical field quantization and scalar fields; visibility derivatives and discontinuity-aware estimators. Exit: Scalar-field quantization derivation plus visibility-gradient test.

Weeks 7-9: A16-2 - QFT foundations + inverse graphics II; Spinors/gauge fields + joint inverse problems. Dirac fields, gauge symmetry and propagators; joint recovery of geometry/material/lighting with constraints. Exit: Spinor/gauge calculations plus joint inverse scene experiment.

Weeks 10-12: A16-3 - QFT foundations + inverse graphics II; Scattering/path integrals + uncertainty/regularization. Perturbative scattering and path-integral machinery; regularization, posterior/uncertainty viewpoints and adversarial inverse cases. Exit: Scattering calculation plus uncertainty-aware inverse-graphics study.

Y11 Q3 (global Q43) - 480 h

Applied/research intensive - 40 h/week (480 h): four consecutive 3-week, 120-hour segments. Treat each segment as a short full-time course: Week A theory/derivations, Week B problems/implementation, Week C validation/reproduction/oral gate.

Weeks 1-3: A17-1 - QFT/QED/renormalization + neural rendering I; QED perturbation theory + NeRF fundamentals. QED rules/amplitudes and regularization setup; positional encoding, volume rendering and neural-field training. Exit: Tree-level QED calculation plus small NeRF reproduction.

Weeks 4-6: A17-2 - QFT/QED/renormalization + neural rendering I; Loops/renormalization + accelerated neural fields. One-loop structure, counterterms and RG competence; multiresolution/hash/grid acceleration and sampling. Exit: Renormalization exercise plus accelerated neural-field benchmark.

Weeks 7-9: A17-3 - QFT/QED/renormalization + neural rendering I; QFT synthesis + representation diagnostics. Ward identities/consistency and renormalized observables; geometry/appearance entanglement, generalization and controlled scene diagnostics. Exit: QFT consistency audit plus neural-representation error decomposition.

Weeks 10-12: A18-1 - Statistical field theory/RG + neural rendering II; Wilsonian RG + Gaussian splatting. Coarse graining, relevant/marginal/irrelevant operators and fixed points; 3D Gaussian splatting representation and rasterization. Exit: RG flow calculation plus 3DGS reproduction.

Y11 Q4 (global Q44) - 480 h

Applied/research intensive - 40 h/week (480 h): four consecutive 3-week, 120-hour segments. Treat each segment as a short full-time course: Week A theory/derivations, Week B problems/implementation, Week C validation/reproduction/oral gate.

Weeks 1-3: A18-2 - Statistical field theory/RG + neural rendering II; Critical phenomena + dynamic neural scenes. Scaling, universality, critical exponents and correlation functions; temporal/deformable scene representations. Exit: Critical-scaling analysis plus dynamic-scene benchmark.

Weeks 4-6: A18-3 - Statistical field theory/RG + neural rendering II; Statistical field theory + editing/relighting. Landau-Ginzburg/statistical fields and perturbative RG entry; editable/relightable neural scenes and disentanglement tests. Exit: Field-theory synthesis plus controlled edit/relight evaluation.

Weeks 7-9: A19-1 - Kinetic/non-equilibrium physics I + generative 3D I; Boltzmann/BBGKY + generative geometry. Kinetic hierarchy, collision operators and conservation; generative point/mesh/implicit geometry models. Exit: Kinetic derivation plus geometry-generation baseline.

Weeks 10-12: A19-2 - Kinetic/non-equilibrium physics I + generative 3D I; Langevin/Fokker-Planck + generative appearance/scenes. Stochastic dynamics and Fokker-Planck equations; diffusion/score or autoregressive scene/appearance models. Exit: SDE/Fokker-Planck calculation plus generative scene experiment.

Year-end flexible weeks (Weeks 49-52). Research defense, repository freeze, literature map refresh, publication-quality animation/figures, and recovery. New topics enter only if they directly serve the capstone.

Year 12: full-time applied and research synthesis - 1,920 scheduled hours

Y12 Q1 (global Q45) - 480 h

Applied/research intensive - 40 h/week (480 h): four consecutive 3-week, 120-hour segments. Treat each segment as a short full-time course: Week A theory/derivations, Week B problems/implementation, Week C validation/reproduction/oral gate.

Weeks 1-3: A19-3 - Kinetic/non-equilibrium physics I + generative 3D I; Linear response + evaluation. Kubo/response and fluctuation-dissipation ideas; distributional, geometric and perceptual evaluation with failure taxonomy. Exit: Response derivation plus statistically disciplined generative-model evaluation.

Weeks 4-6: A20-1 - Hydrodynamic/response synthesis + generative 3D II; Hydrodynamic limits + generative 3D constraints. Scaling from kinetic to fluid descriptions; impose geometry, topology or physical constraints in generation. Exit: Limit-scaling note plus constrained generation experiment.

Weeks 7-9: A20-2 - Hydrodynamic/response synthesis + generative 3D II; Nonequilibrium synthesis + 4D generation. Entropy production, steady states and transport; dynamic/4D scene or motion generation. Exit: Nonequilibrium problem set plus 4D generation baseline.

Weeks 10-12: A20-3 - Hydrodynamic/response synthesis + generative 3D II; P19 bridge + rigorous evaluation. Connect response/kinetics to quantum matter; evaluate physical consistency, temporal coherence and out-of-distribution behavior. Exit: Cross-physics synthesis plus adversarial 3D/4D benchmark.

Y12 Q2 (global Q46) - 480 h

Applied/research intensive - 40 h/week (480 h): four consecutive 3-week, 120-hour segments. Treat each segment as a short full-time course: Week A theory/derivations, Week B problems/implementation, Week C validation/reproduction/oral gate.

Weeks 1-3: A21-1 - Advanced quantum matter I + visual computing breadth; Topological quantum matter + scientific visualization. Berry phase/curvature, Chern-type invariants and edge/bulk ideas; scalar/vector/tensor visualization and uncertainty. Exit: Topological-band calculation plus scientific visualization project.

Weeks 4-6: A21-2 - Advanced quantum matter I + visual computing breadth; Strong correlation + interaction/VR. Model Hamiltonians, Mott/collective phenomena and numerical intuition; interaction design, tracking, latency and VR/AR systems. Exit: Strong-correlation study plus immersive prototype/latency analysis.

Weeks 7-9: A21-3 - Advanced quantum matter I + visual computing breadth; Quantum criticality + fabrication. Quantum phase transitions and scaling; computational fabrication, toolpaths/process constraints and graphics-to-physical pipelines. Exit: Criticality synthesis plus fabrication-aware geometry pipeline.

Weeks 10-12: A22-1 - Advanced quantum matter II/EFT + mathematical animation; Effective field theory + mathematical animation language. Power counting, integrating out scales and operator bases; advanced Manim architecture, camera/time control and reusable visual primitives. Exit: EFT derivation plus polished theorem/physics animation.

Y12 Q3 (global Q47) - 480 h

Applied/research intensive - 40 h/week (480 h): four consecutive 3-week, 120-hour segments. Treat each segment as a short full-time course: Week A theory/derivations, Week B problems/implementation, Week C validation/reproduction/oral gate.

Weeks 1-3: A22-2 - Advanced quantum matter II/EFT + mathematical animation; Advanced quantum matter + visual proofs. Topological/response/effective descriptions; animate proofs, geometric constructions and dynamical systems without hiding logical steps. Exit: Research-level visual proof with accompanying rigorous text.

Weeks 4-6: A22-3 - Advanced quantum matter II/EFT + mathematical animation; Matter synthesis + simulation explanation. Connect symmetry, topology and emergent descriptions; build explanatory animations for PDE/simulation/quantum phenomena with quantitative overlays. Exit: 10-15 minute research-grade animated lecture or equivalent scene suite.

Weeks 7-9: A23-1 - Advanced quantum fields + graphics apprenticeship I; Gauge/anomaly/topological QFT + reproduction protocol. Anomalies, gauge consistency and topological terms; paper triage, environment capture and baseline reproduction. Exit: Mathematical-physics derivation plus reproducible graphics-paper baseline.

Weeks 10-12: A23-2 - Advanced quantum fields + graphics apprenticeship I; Nonperturbative/advanced fields + benchmarking. Instanton/soliton/nonperturbative or specialist structures; benchmark design, ablations and adversarial cases. Exit: Advanced-field note plus benchmark suite with uncertainty/error bars.

Y12 Q4 (global Q48) - 480 h

Applied/research intensive - 40 h/week (480 h): four consecutive 3-week, 120-hour segments. Treat each segment as a short full-time course: Week A theory/derivations, Week B problems/implementation, Week C validation/reproduction/oral gate.

Weeks 1-3: A23-3 - Advanced quantum fields + graphics apprenticeship I; Mathematical physics synthesis + research question. Connect topology/geometry/field theory to specialist papers; turn reproduction failures into falsifiable graphics research questions. Exit: Research memo containing literature map, reproduced result and proposed experiment.

Weeks 4-6: A24-1 - Joint research synthesis; Portfolio convergence. Select one mathematical, one physical and one graphics research thread; reconcile notation, assumptions, code and evidence; identify a unifying question. Exit: Unified 10-20 page prospectus and clean research repository.

Weeks 7-9: A24-2 - Joint research synthesis; Independent reproduction / original extension. Reproduce one technically demanding paper or theorem-assisted computation, then add a nontrivial extension, stronger test or new regime. Exit: Reproduction report with exact deviations, ablations and extension result.

Weeks 10-12: A24-3 - Joint research synthesis; Capstone defense. Write research-style final document, archive proofs/code/data/animations, conduct oral defense and formulate the next 12-month specialist agenda. Exit: Defensible cross-domain capstone plus publication/research roadmap.

Year-end flexible weeks (Weeks 49-52). Research defense, repository freeze, literature map refresh, publication-quality animation/figures, and recovery. New topics enter only if they directly serve the capstone.

4.1 Twelve-year milestone ladder

Milestone	Competence expected	Evidence
End Year 1	Proof language, linear algebra, algebra and real analysis underway/owned at scheduled level; reproducible scientific-computing environment; first mechanics/image-science bridge started.	Closed-book proof exam + tested numerical library + animation portfolio.
End Year 2	Logic/homological foundations, complex analysis and topology; mechanics and real-time graphics substantially developed.	Cumulative theorem map + renderer/animation artifacts.
End Year 3	Probability, functional analysis, manifolds and elementary/analytic number theory; electromagnetism/ray-tracing bridge active.	Operator/probability exam + validated ray-tracing/EM projects.
End Year 4	Harmonic analysis, PDE, Lie/representation theory and algebraic number theory; EM/SR, light transport and statistical mechanics linked rigorously.	PDE/harmonic proof gate + path-tracing/stat-mech diagnostics.
End Year 5	Diophantine approximation, SDE/random structures, smooth dynamics and geometric group theory; quantum mechanics/geometric modeling underway.	Research-style math note + QM/geometry code portfolio.
End Year 6	Additive combinatorics, expanders, advanced analytic NT and finite fields; QM II/DDG and continuum/fluid bridge.	Spectral-gap/additive synthesis + DDG/fluid validation project.
End Year 7	Diophantine geometry/transcendence, automorphic forms, algebraic geometry and arithmetic geometry; GR/simulation and condensed matter bridge.	Arithmetic proof defense + GR/FEM simulation evidence.
End Year 8	Cryptography, percolation/SLE, particle systems/Ising and homogeneous dynamics; semiconductor/simulation II bridge.	Probability/dynamics research memo + device/fluid/MPM code.
End Year 9	Fractal/entropy methods, modern harmonic/geometric analysis, geometric topology/Ricci flow, etale/Weil-Deligne; many-body/computational imaging begins.	Advanced theorem architecture defense + inverse-imaging project.
End Year 10	Four mathematics synthesis/capstone quarters complete; A14-I complete.	Mathematics research capstone defended; no remaining foundational mathematics coursework.
End Year 11	Many-body II, inverse graphics, QFT, neural rendering, RG and kinetic/generative work at intensive research pace.	At least two paper reproductions with adversarial evaluation.
End Year 12	Advanced quantum matter/fields, visualization/fabrication, mathematical animation and G18/P20 joint capstone complete.	Research-style thesis/capstone, reproducible repository, oral defense and next-specialization plan.

4.2 Quarter advancement rule

- Mathematics gate: no advancement with unresolved failure on a foundational theorem, definition cluster or core problem family. Reallocate part or all of the next quarter's 10 applied hours to repair if necessary.
- Applied gate: a project is not passed because it runs. It must have a stated model, unit/limit checks, numerical or statistical diagnostics, reproducible code and a short account of observed failure modes.
- Reading is never counted as mastery evidence by itself. Every quarter must produce proofs/problems and an applied artifact.
- Paper reading becomes a larger fraction only after the relevant textbook-level derivations and reference implementations are secure.
- The four flexible weeks each year are buffers and audits, not hidden fifth quarters. Chronic spillover means the schedule should be slowed rather than the rigor standard lowered.

5. Mathematics library: the rigorous spine

All M-volumes are retained. The short descriptions below are an integration index; execute the detailed v6 unit schedules, exact readings and precision problems in the mathematics source plan.

M01. Proof and Mathematical Language

Owens. Propositional/quantified logic, sets, functions, proof methods, induction, construction and counterexample discipline.

Feeds. Proof literacy for every later volume; physics assumptions and graphics correctness arguments.

M02. Linear Algebra

Owens. Vector spaces, linear maps, duality, determinants, inner products, spectral theory, bilinear/quadratic forms and canonical decompositions.

Feeds. Quantum states/operators, transformations, least squares, geometry, numerical linear algebra.

M03. Abstract Algebra

Owens. Groups, rings, fields, modules, group actions, quotient structures, polynomial algebra and Galois-theoretic entry.

Feeds. Symmetry, representation theory, finite fields, arithmetic and topology.

M04. Commutative and Homological Algebra

Owens. Modules, localization, Noetherian ideas, exact sequences, homological language and the algebra supporting schemes/cohomology.

Feeds. Algebraic geometry, arithmetic geometry, topology and modern representation/arithmetic tools.

M05. Mathematical Logic

Owens. First-order logic, formal theories, completeness/compactness, computability, Godel incompleteness, set theory and independence themes including CH.

Feeds. Clarifies proof, definability, decidability and Hilbert tenth; no simplified duplicate elsewhere.

M06. Real Analysis

Owens. Sequences/functions, metric spaces, compactness, differentiation/integration, uniform convergence and rigorous multivariable analytic foundations.

Feeds. Mechanics, PDE, probability, numerical analysis and graphics continuous models.

M07. Functional Analysis

Owens. Normed/Banach/Hilbert spaces, bounded operators, Hahn-Banach, uniform boundedness, open/closed graph principles and spectral/operator theory.

Feeds. Quantum mechanics/QFT, PDE, inverse problems and rigorous optimization/operator language.

M08. Complex Analysis

Owens. Holomorphic functions, Cauchy theory, residues, harmonic/conformal methods and analytic continuation.

Feeds. EM/QM contour methods, analytic number theory, SLE/conformal theory.

M09. Harmonic Analysis

Owens. Fourier transforms/series at theorem level, distributions as needed, singular integrals, interpolation and harmonic-analytic operator methods.

Feeds. PDE, rendering sampling, signal/image analysis, Carleson/decoupling endpoint.

M10. Partial Differential Equations

Owens. Weak formulations, Sobolev spaces, elliptic/parabolic/hyperbolic theory, variational methods, regularity and core nonlinear PDE tools.

Feeds. Fluid dynamics, field equations, GR, kinetic limits and graphics simulation.

M11. Topology

Owens. Point-set topology and algebraic-topological foundations: homotopy/fundamental group, homology/cohomology structures and exactness.

Feeds. Manifolds, gauge/GR topology, geometric topology and shape processing.

M12. Smooth Manifolds and Differential Geometry

Owens. Manifolds, tangent/cotangent bundles, tensors, differential forms, Stokes, Riemannian geometry and curvature foundations.

Feeds. Hamiltonian mechanics, GR, Lie groups, continuum geometry and DDG.

M13. Geometric Topology, Ricci Flow and Hauptvermutung

Owens. Manifold/3-manifold topology, Ricci-flow proof architecture for Poincare, triangulation/PL-topological issues around Hauptvermutung.

Feeds. Geometric PDE and topology capstone; informs shape topology but is not reduced to computation.

M14. Measure-Theoretic Probability

Owens. Probability spaces, conditional expectation, martingales, convergence modes, limit theorems, concentration and rigorous stochastic foundations.

Feeds. Statistical mechanics, Monte Carlo rendering, stochastic physics and random structures.

M15. SDEs and Random Structures

Owens. Brownian motion, Ito calculus, SDEs, stochastic processes, random graphs/matrices and probabilistic structures.

Feeds. SLE, nonequilibrium physics, stochastic numerics and random-media models.

M16. Percolation and SLE

Owens. Critical percolation, planar methods, Loewner evolution, SLE martingales and rigorous scaling-limit architecture.

Feeds. Modern probability/mathematical physics research.

M17. Particle Systems, Kinetic Limits and Ising

Owens. Interacting particles, BBGKY/propagation of chaos, kinetic/hydrodynamic limits, Gibbs/Ising methods and modern rigorous endpoints.

Feeds. Rigorous counterpart to P10/P18; includes Hilbert-sixth and hard-sphere research pathways.

M18. Lie Groups and Representation Theory

Owens. Lie groups/algebras, semisimple structure, homogeneous spaces, finite/continuous/unitary representations and key representation-theoretic tools.

Feeds. QM/QFT symmetry, homogeneous dynamics, property (T), automorphic theory.

M19. Smooth Dynamics, KAM and Billiards

Owens. Dynamical systems, hyperbolicity/ergodicity foundations, smooth dynamics, KAM small-divisor methods and billiards.

Feeds. Classical mechanics refinement, rigidity and nonlinear dynamics.

M20. Homogeneous Dynamics

Owens. Lattices, unipotent/diagonal dynamics, Ratner theory, effective equidistribution, EKL, Marklof-Strombergsson, Benoist-Quint and Diophantine applications.

Feeds. High-level dynamics/number theory research track.

M21. Geometric Group Theory and Property (T)

Owens. Cayley geometry, quasi-isometry, hyperbolicity, amenability/random walks, Kazhdan property (T)/(tau) and expander connections.

Feeds. Rigidity, spectral gaps, Zimmer and arithmetic group actions.

M22. Elementary and Analytic Number Theory

Owens. Elementary arithmetic, primes, Dirichlet series/L-functions, PNT/Dirichlet-type architecture and analytic number-theory foundations.

Feeds. Gateway to all arithmetic research volumes.

M23. Advanced Analytic Number Theory

Owens. Sieve/circle/exponential-sum methods, advanced L-function techniques and research-facing analytic tools.

Feeds. Decoupling interfaces, primes/arithmetic applications, automorphic/arithmetic synthesis.

M24. Algebraic Number Theory

Owens. Number fields, Dedekind domains, ideals, valuations/completions, local fields, class groups/units and class-field-theoretic direction.

Feeds. Diophantine geometry, automorphic forms, arithmetic geometry and cryptography.

M25. Geometry of Numbers and Diophantine Approximation

Owens. Lattices, successive minima, Minkowski methods, continued fractions/transference, metric and homogeneous viewpoints.

Feeds. Baker/Schmidt paths, dynamics, fractal Diophantine approximation.

M26. Diophantine Geometry, Transcendence and Hilbert 10

Owens. Heights and Diophantine equations; Baker linear forms in logarithms, Schmidt Subspace Theorem, S-unit themes, undecidability/Hilbert tenth architecture.

Feeds. Arithmetic research and transcendence/definability synthesis.

M27. Finite Fields

Owens. Finite-field structure, characters, Gauss/Jacobi sums, curves/point counting and arithmetic/combinatorial tools.

Feeds. Weil theory, coding/crypto, expanders and additive combinatorics.

M28. Additive Combinatorics and Approximate Groups

Owens. Ruzsa calculus, energy, BSG/Freiman, sum-product, Gowers-type structure and approximate-group methods.

Feeds. Bourgain-Gamburd, expansion, harmonic/arithmetic combinatorics.

M29. Expanders, Spectral Gaps and Bourgain-Gamburd

Owens. Spectral graph theory, expansion, property (τ), product growth, flattening and Bourgain-Gamburd style spectral-gap mechanisms.

Feeds. Arithmetic expansion, random walks and rigidity.

M30. Automorphic Forms

Owens. Modular/automorphic forms, representation/adeles perspective, Hecke operators, L-functions and automorphy architecture.

Feeds. FLT, Sato-Tate and modern arithmetic geometry.

M31. Algebraic Geometry

Owens. Varieties/schemes, sheaves, morphisms, divisors, cohomological language and core scheme-theoretic geometry.

Feeds. Arithmetic geometry, étale cohomology and moduli viewpoints.

M32. Arithmetic Geometry

Owens. Elliptic/abelian varieties, heights, Galois representations, Faltings/Mordell, modularity/FLT and Sato-Tate proof architecture.

Feeds. Flagship arithmetic capstones.

M33. Etale Cohomology and Weil/Deligne

Owens. Etale topology/cohomology, l-adic methods, Frobenius, Weil conjectures and Deligne/Bombieri architecture over finite fields.

Feeds. Deep arithmetic-geometric endpoint.

M34. Fractal Measures and Entropy Methods

Owens. Self-similar measures, dimension/entropy, Hochman inverse entropy, Shmerkin inverse L_q ideas, Bernoulli convolutions and Varju-type arithmetic/fractal methods.

Feeds. Fractal Diophantine problems, dynamics and modern measure theory.

M35. Modern Harmonic and Geometric Analysis

Owens. Carleson, restriction/decoupling, wave packets, polynomial partitioning, Keakeya/Falconer/Furstenberg and modern Hong-Wang-type geometric analysis.

Feeds. Research endpoint for analysis/geometric measure theory.

M36. Cryptography

Owens. Rigorous public-key primitives, finite-field/elliptic arithmetic, security notions/reductions and computational number-theory interfaces.

Feeds. Applied arithmetic track; mathematical correctness and security definitions both matter.

6. Physics library after mathematical deduplication

Physics Volumes I-V are replaced by M-volumes. The fifteen retained physics volumes below are studied for physical content, not as second mathematics courses. When a physics text states a theorem informally, consult the owning M-volume for the rigorous version.

P06. Classical Mechanics

Physics content. Newtonian, Lagrangian and Hamiltonian mechanics; variational principles, constraints, oscillations, canonical structure and nonlinear dynamics.

Mathematical support. M02, M06; later M12/M19 for geometry and dynamics.

P07. Electromagnetism and Special Relativity

Physics content. Electrostatics/magnetostatics, Maxwell equations, waves/radiation, covariant electrodynamics and special relativity.

Mathematical support. M06, M08, M09, M10; supplies physical optics to G06/G12.

P08. Quantum Mechanics I

Physics content. Hilbert-space states/operators, measurement, dynamics, symmetries, spin, angular momentum, hydrogen and entanglement.

Mathematical support. M02, M07, M18; operator-domain rigor is upgraded, not bypassed.

P09. Quantum Mechanics II

Physics content. Perturbation/variational/WKB methods, scattering, partial waves, density matrices/open systems, path integrals and relativistic preview.

Mathematical support. M07/M18 plus asymptotic methods retained here as applied mathematics.

P10. Thermodynamics and Statistical Mechanics

Physics content. Thermodynamic structure, ensembles, partition functions, ideal/interacting systems, phase transitions and statistical-mechanical reasoning.

Mathematical support. M14 probability; rigorous Ising/particle endpoints later in M17.

P11. Continuum Mechanics and Fluid Dynamics

Physics content. Continuum kinematics, stress/strain, conservation laws, Euler/Navier-Stokes, viscosity, shocks, stability and turbulence.

Mathematical support. M10 owns PDE rigor; G10-G11 own discretization/simulation.

P12. Condensed Matter Physics I

Physics content. Crystals, reciprocal lattices, phonons, electrons in solids, band theory, semiclassical dynamics and basic transport.

Mathematical support. M02/M07/M09; precedes semiconductors and many-body theory.

P13. Condensed Matter Physics II

Physics content. Second quantization, Hartree-Fock, Green functions, quasiparticles, collective modes, superconductivity, superfluidity, correlation and emergent phases.

Mathematical support. M07/M18 plus P08/P10/P12.

P14. Semiconductor Physics and Quantum Electronic Devices

Physics content. Carrier statistics/transport, Poisson-continuity device PDEs, junctions, MOS, heterostructures, quantum wells and Landauer/NEGF transport.

Mathematical support. P08/P12 plus M10/M14; computational device models are retained.

P15. Classical Field Theory and General Relativity

Physics content. Field variational calculus, tensors/connections/curvature, Einstein equations, exact solutions, causal/geometric applications.

Mathematical support. M12/M18/M10; topology from M11 as needed.

P16. Quantum Field Theory

Physics content. Classical fields to quantization, scalar/spinor/gauge fields, perturbative scattering, path integrals, QED and renormalization.

Mathematical support. M07/M09/M10/M12/M18 plus P09/P15.

P17. Renormalization, Critical Phenomena and Statistical Field Theory

Physics content. Wilsonian RG, universality, critical exponents, field-theoretic stat mech and scaling.

Mathematical support. P10/P16 plus M09/M10/M14.

P18. Kinetic Theory, Stochastic Physics and Nonequilibrium Statistical Mechanics

Physics content. Boltzmann/BBGKY viewpoint, Langevin/Fokker-Planck, linear response, fluctuation-dissipation, hydrodynamic/kinetic limits and nonequilibrium structure.

Mathematical support. M14/M15/M10; M17 provides the rigorous particle-limit research counterpart.

P19. Advanced Quantum Matter

Physics content. Strong correlation, topological phases, quantum criticality, advanced response and effective-field viewpoints in matter.

Mathematical support. P13/P17/P18 plus M18 as symmetry language.

P20. Advanced Quantum Fields and Mathematical Physics

Physics content. EFT, gauge/anomaly/topological structures, advanced field-theory methods and specialist research-paper entry.

Mathematical support. P16-P19 plus M07/M12/M18 and relevant topology/geometry.

7. Computer graphics library after mathematical deduplication

Graphics Volume 1 is dissolved. Abstract mathematics moves to the M-spine; its unique numerical/optimization material is retained in an expanded G02. Every later graphics volume keeps the v4 research cycle: derive -> implement -> diagnose -> compare -> reproduce -> formulate a research question.

G02. Scientific Computing, Optimization, Programming and GPU Systems

Graphics content. Modern Python/C++, numerical conditioning, sparse/numerical linear algebra, least squares/optimization, autodiff, time integrators, testing/profiling, GPU execution and parallel primitives. This absorbs the non-duplicate numerical core of old G01.

Cross-domain rule. M02/M06; numerical practice remains here even when theory is known.

G03. Images, Sampling, Color and Visual Perception

Graphics content. Image representation, sampling/reconstruction, antialiasing, color science, tone/HDR and perception-aware image reasoning.

Cross-domain rule. M09 owns Fourier theorem rigor; G03 owns sampled-image consequences and visual systems.

G04. Rasterization and Real-Time Rendering

Graphics content. Raster pipeline, barycentrics, clipping, interpolation, textures/filtering, shading, visibility/shadows, PBR approximations and GPU real-time effects.

Cross-domain rule. M02 plus G02/G03.

G05. Ray Tracing and Rendering Systems

Graphics content. Ray intersections, acceleration structures/BVH, cameras, materials, recursive transport and robust renderer architecture.

Cross-domain rule. M02/M06 plus G02.

G06. Physically Based Rendering and Light Transport

Graphics content. Radiometry, BSDFs, rendering equation, Monte Carlo estimators, path tracing, MIS, participating media and advanced path-space methods.

Cross-domain rule. P07 supplies wave/optical physics where needed; M14 owns probability foundations.

G07. Geometric Modeling, CAGD and Shape Representations

Graphics content. Meshes, Bezier/B-spline/NURBS, subdivision, implicit/SDF/CSG representations and modeling operations.

Cross-domain rule. M02/M12; keep representation/algorithm specifics.

G08. Geometry Processing and Discrete Differential Geometry

Graphics content. Discrete operators, curvature, Laplacians, parameterization, geodesics, deformation, remeshing, topology/robustness and vector-field/Hodge tools.

Cross-domain rule. M12 supplies smooth geometry; M10 supplies variational/PDE foundations.

G09. Animation, Kinematics and Character Motion

Graphics content. Rigid transforms, quaternions, interpolation, skeletal animation, inverse kinematics, constraints and motion/control representations.

Cross-domain rule. P06 mechanics and M02 linear algebra; this volume is non-physical motion representation before full simulation.

G10. Physics-Based Simulation I

Graphics content. Particles, rigid bodies, collision/contact, deformable solids, FEM, cloth, shells/rods and robust time stepping.

Cross-domain rule. P06/P11 supply physical models; M10 supplies PDE/variational rigor; G10 owns discretization/contact/solver diagnostics.

G11. Physics-Based Simulation II

Graphics content. Eulerian fluids, pressure projection, PIC/FLIP/APIC, MPM, free surfaces, multiphysics and hybrid continuum simulation.

Cross-domain rule. P11 and M10 are prerequisites; convergence/conservation testing is mandatory.

G12. Cameras, Optics, Computational Imaging and Displays

Graphics content. Camera/sensor/optics models, image formation, computational photography, inverse imaging and immersive/display systems.

Cross-domain rule. P07 owns EM/wave foundations; G12 owns sensing/computation and inverse-imaging systems.

G13. Differentiable Rendering and Inverse Graphics

Graphics content. Differentiation through image formation, visibility discontinuities, inverse estimation of shape/material/light/pose, regularization and identifiability.

Cross-domain rule. M07/M14 plus G06/G12; numerical optimization from G02.

G14. Neural Rendering and Neural Scene Representations

Graphics content. Implicit neural fields/NeRF, acceleration, Gaussian splatting, dynamic/editable/relightable scenes and representation diagnostics.

Cross-domain rule. G13 plus probability/optimization foundations; evaluate representation error separately from training error.

G15. Generative 3D/4D and Machine Learning for Graphics

Graphics content. Generative geometry/appearance/scenes/motion, diffusion and learned priors, physical/geometric constraints and evaluation of 3D/4D generation.

Cross-domain rule. G07/G09/G14; preserve geometric/physical validity rather than treating output plausibility as correctness.

G16. Visualization, Interaction, VR/AR, Fabrication and Special Topics

Graphics content. Scientific/information visualization, interaction, immersive systems, fabrication and graphics-to-physical pipelines.

Cross-domain rule. Broad synthesis volume with domain-specific human/system constraints.

G17. Mathematical Animation and Visual Explanation

Graphics content. Manim/programmatic animation, geometric transformations, camera/time control, visual proofs, algorithm/simulation animation and research communication.

Cross-domain rule. Parallel hobby/research communication track; use to visualize M/P/G concepts without replacing proof.

G18. Research Apprenticeship, Paper Reproduction and Open Problems

Graphics content. Paper triage, reproduction, benchmarks, adversarial evaluation, ablations, literature maps, research question formation and capstone execution.

Cross-domain rule. Begins informally after G06 and becomes dominant at the end.

8. High-value cross-domain bridges

Bridge	Route	What to do
Fourier analysis -> sampling -> imaging	M09 -> G03/G12 -> P07	Prove Fourier results in M09, then study aliasing/filtering and computational imaging as consequences. In physics, connect frequency-domain methods to waves and response.
Variational calculus -> mechanics -> simulation	M06/M10/M12 -> P06/P11 -> G10/G11	Derive Euler-Lagrange/weak formulations rigorously, interpret energy and conservation physically, then discretize and test convergence/stability.
Probability -> Monte Carlo -> statistical physics	M14/M15 -> G06 -> P10/P18	The same expectation/concentration/martingale language supports unbiased transport estimators and ensembles/non-equilibrium processes.
Differential geometry -> GR -> DDG	M12 -> P15 -> G08	Learn tensors/curvature smoothly, apply them to spacetime, then study what discrete mesh operators preserve or fail to preserve.
Functional analysis -> QM/QFT -> inverse graphics	M07 -> P08/P16 -> G13	Hilbert/operator structure prevents sloppy quantum foundations; inverse graphics uses the same operator/regularization viewpoint in finite/infinite-dimensional settings.
PDE theory -> fluids -> graphics fluids	M10 -> P11 -> G11	Separate existence/regularity/weak-solution questions from physical closure and from numerical scheme behavior.
Lie/representation theory -> symmetry	M18 -> P08/P16/P15	Use exact group/representation language for angular momentum, gauge/Lorentz symmetries and field representations.
Statistical mechanics -> rigorous particle limits	P10/P18 -> M17	Physics gives the model hierarchy and intuition; M17 supplies modern proof frameworks from particle systems to kinetic/field equations and rigorous Ising theory.
Fractal/harmonic analysis -> modern scene representations	M34/M35 -> G14/G15	Use multiscale, measure and approximation ideas as mathematical lenses for neural/point-based scene representations rather than as superficial analogies.
Visualization as a proof/debugging instrument	G17 -> all tracks	Animate only after identifying invariants and claims. A visualization may reveal geometry or failure modes, but it never substitutes for a theorem proof or quantitative validation.

8.1 One notebook system, three kinds of evidence

Artifact	Required evidence
Mathematics theorem notebook	Exact statements/hypotheses; counterexamples; full proofs or proof architectures; dependency arrows; re-proofs after delay.
Physics derivation/model notebook	Governing principles; scaling; approximation parameters; units; limiting cases; physical interpretation; comparison with numerics/experiment where available.
Graphics research repository	Reference implementation; unit/regression/adversarial tests; configs/seeds; convergence/stability/runtime plots; paper reproductions; failure ledger.

9. Mastery gates: when you may move on

Gate	Pass evidence
Mathematical object gate	Define every central object with quantifiers/ambient structure; give examples and near-misses; explain invariance/functorial behavior where relevant.
Theorem gate	For foundations, reproduce the proof. For frontier theorems, state precisely, draw the dependency graph, reconstruct at least one key lemma, and know why hypotheses matter.
Physics gate	Derive the canonical relation/equation; state units, symmetry, approximation regime and limiting cases; solve an unfamiliar problem without being told the method.
Discretization gate	Derive or justify the discrete scheme from the continuous model; identify consistency/stability/conservation properties and mesh/timestep dependence.
Implementation gate	Pass ordinary, boundary and adversarial regression cases; compare against an analytic, hand-worked or refined numerical oracle.
Research gate	Reproduce a nontrivial paper component, run a controlled alternative or ablation, identify the strongest alternative explanation, and formulate a falsifiable next question.
Communication gate	Give a board explanation and, where useful, a mathematical animation that preserves the actual invariants rather than merely looking plausible.

9.1 Failure policy

A failed checkpoint is not repaired by reading more pages. Diagnose the failure class and return to the smallest owning volume.

Failure signature	Repair route
Undefined object / theorem misuse	Return to the M-volume owning the definition/theorem.
Correct algebra, wrong physical conclusion	Return to the P-volume assumptions, scale analysis and limiting regime.
Visually good, quantitatively wrong simulation	Return to G02/G10/G11: discretization, conditioning, timestep, constraint or solver diagnostics.
Inverse/neural model fits training but fails transfer	Separate identifiability, representation capacity, optimization, data coverage and estimator noise; use G13-G15 research tests.
Repeated inability to reconstruct after 48-72 h	Reduce new material and increase closed-book proof/problem retrieval; do not advance the calendar.

10. Research transition and capstones

The unified programme should eventually stop behaving like three curricula and become one research environment. The final years therefore use synthesis quarters and paired capstones rather than continued breadth accumulation.

Research family	Primary dependency bundle	Illustrative endpoint
Arithmetic / dynamics / fractals	M25-M34 + MS37/MS38	Examples: homogeneous dynamics and Diophantine approximation; spectral gaps/approximate groups; fractal measures/entropy; arithmetic geometry.
Harmonic / geometric analysis	M09/M23/M35 + MS39	Examples: Carleson, decoupling, Kakeya/Falconer/Furstenberg, polynomial partitioning, restriction and incidence geometry.
Mathematical physics / kinetic limits	M10/M14-M17 + P10/P11/P18	Examples: particle-to-kinetic/hydrodynamic limits, Ising/critical systems, rigorous nonequilibrium questions.
Quantum matter / fields	M07/M12/M18 + P13/P16-P20	Examples: many-body/field-theoretic structures, EFT, topology/anomalies and advanced mathematical physics.
Rendering / inverse graphics	M09/M14 + P07 + G05/G06/G12-G14	Examples: light transport estimators, differentiable rendering, inverse imaging, neural scene representations and identifiability.
Geometry / simulation	M10/M12 + P06/P11 + G07-G11	Examples: discrete geometry, contact/FEM/MPM, conservative or structure-preserving simulation, geometry processing.
Generative 3D / visual computing	G14-G18 with relevant M/P prerequisites	Examples: structured 3D generation, physically constrained generation, robust evaluation, visualization and fabrication.

10.1 Final portfolio standard

- Four seminar talks: at least two mathematics-heavy, one physics-heavy, and one graphics/computational.
- One 40-60 page expository manuscript with complete definitions, precise theorem statements, a dependency map, and selected reconstructed proofs/derivations.
- At least two serious paper reproductions: one computational/graphics or computational-physics result and one proof-oriented mathematical result or lemma chain.
- One integrated project in which a rigorous mathematical structure materially changes a physics or graphics method, evaluation, or interpretation.
- A living research map: known results, standard examples/counterexamples, unresolved bottlenecks, benchmark weaknesses and 3-5 tractable subproblems.

10.2 Suggested integrated capstone patterns

Pattern	Minimum standard
Structure-preserving simulation	Start from M10/M12 variational or geometric structure, formulate a P11 physical model, implement in G10/G11, and test whether the discrete method preserves the predicted invariant under refinement.
Inverse light transport with identifiability	Use M07/M14 inverse/operator/probability language, P07 optical constraints and G06/G13 differentiable rendering; prove/diagnose which parameters are identifiable and which are confounded.
Kinetic limit as a visualization/computation project	Use M17 rigorous limit architecture and P18 kinetic theory, then build G17/G16 visualizations or validated simulations that expose scaling regimes without conflating numerical evidence with proof.
Geometric analysis for shape processing	Use M12/M35 geometric/analytic ideas and G08 mesh algorithms; test discrete convergence on analytic surfaces and adversarial meshes, then formulate a research question around stability/robustness.
Neural scene representation through approximation theory	Use M09/M34 multiscale/measure ideas to design diagnostics for G14 representations; distinguish approximation, sampling and optimization effects with controlled synthetic scenes.

11. How to use the three detailed PDFs with this master plan

This document is the integration and scheduling layer. It intentionally does not reprint the hundreds of already-completed week cards, exact chapter locators and problem ladders. Use the detailed PDFs as execution manuals, but apply the following precedence rules so you do not accidentally reintroduce duplication.

1. When a mathematics component appears in Years 1-10: follow the Mathematics v6 quarter exactly at 30 h/week, including its proof/use standard and unit-specific precision problems. The extra 10 h/week is separate applied capacity.
2. When a P-volume appears: follow the Refined Physics topic cards by allocated hours rather than by their original calendar-week numbering; skip foundational mathematics already certified by the owning M-volume and replace informal statements with the M-version.
3. When a G-volume appears: follow Graphics v4 implementation/research tasks. For old G01 references, use M-volumes for pure mathematics and expanded G02 for numerical/optimization practice.
4. Do not delete a physics derivation merely because the governing equation is mathematically known: deriving it from physical principles and understanding the approximation regime is a different competency.
5. Do not delete a graphics implementation merely because the mathematics is known: discretization, robustness, systems behavior and empirical failure analysis are different competencies.
6. If a source-plan schedule conflicts with a prerequisite in this master map, the prerequisite wins. Delay the applied unit; do not dilute the prerequisite.

Precedence rule. Mathematical rigor > prerequisite order > physical derivation/intuition > computational implementation schedule. This is the tie-breaker whenever the three original plans disagree about when or how deeply to introduce a shared concept.

11.1 Annual audit

Audit	Required annual evidence
Proof audit	Randomly select 8 foundational results studied that year and reproduce at least 5 fully closed-book; for the others, write exact failure diagnoses.
Physics audit	Select 6 canonical derivations/models; state assumptions, derive, check dimensions/limits and explain where approximation control enters.
Graphics audit	Rerun two old projects from a clean environment; re-test adversarial cases and verify that reported numbers regenerate without manual editing.
Cross-domain audit	Write one 5-10 page note tracing a single idea across all three layers, e.g. spectral theory, variational principles, Monte Carlo, curvature or multiscale analysis.

11.2 End state

Completion means more than broad familiarity. You should be able to enter a new topic by locating its mathematical owner, derive the physical model when one exists, implement or reproduce the computational method, and decide which failure is mathematical, physical, numerical, statistical or representational. At that point the three disciplines function as one research toolkit rather than three reading lists.

Appendix A. Compact prerequisite crosswalk

Applied block	Mathematical/physical prerequisites	Feeds
P06 Classical mechanics	M02, M06; M12/M19 later	G09-G10
P07 EM/SR	M06, M08-M10	G06, G12
P08-P09 QM	M02, M07, M18	G13 conceptual/operator links
P10 Statistical mechanics	M14-M15	M17 rigorous endpoint; G06 probability analogy
P11 Fluids/continuum	M10, M12	G10-G11
P12-P13 Condensed matter	M02, M07, M09, M14, M18	P14, P19
P14 Semiconductors	P08, P12; M10, M14	Computational device modeling; inverse/design projects
P15 GR	M10-M12, M18	Geometric simulation/visualization links
P16-P17 QFT/RG	M07, M09-M10, M12, M18	P19-P20
P18 Nonequilibrium/kinetic	M14-M15, M10	M17 rigorous particle limits
G02 systems/numerics	M02, M06	All graphics
G03-G06 rendering	M02, M09, M14; P07 for optics	G12-G14
G07-G08 geometry	M02, M12, M10	G09-G11, G13-G15
G09-G11 motion/simulation	P06/P11; M10/M12	Physical animation/simulation research
G12-G13 imaging/inverse	P07; M07, M09, M14	G14-G15
G14-G15 neural/generative	G07/G12/G13; M14	G18 research
G16-G18 breadth/research	All relevant prerequisites	Final portfolio

Appendix B. Source-document provenance

The merge is based on the latest completed versions available in the project file library at the time of integration:

Domain	Source	Role in this merge
Mathematics	Master_Mathematics_30h_Volume_Based_Research_Curriculum_v6_Audited_Specific_Schedules.pdf	10 years, 40 quarters, 36-volume architecture; audited schedule-specific edition.
Physics	Theoretical_Physics_7_Year_Executable_Reading_Curriculum_Refined.pdf	7 years, 28 quarters, 20 volumes, 259 topic weeks; refined tailored problem ladders.
Computer graphics	Computer_Graphics_Research_Master_Curriculum_V4.pdf	18 volumes, 144 modules; concrete programming/research projects and self-contained mastery practicals.

Editorial decision. Future revisions should be made against this ownership map: update the owning volume first, then update cross-domain applications. Do not allow a new physics or graphics revision to quietly recreate a weaker parallel course in mathematics.